



Investigating planar shear layer mixing dynamics via Mori-Zwanzig formalism

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In this work, we apply the data-driven Mori-Zwanzig Mode Decomposition (MZMD), a technique generalizing Dynamic Mode Decomposition (DMD), used for extracting large-scale spatio-temporal coherent structures to analyze the dominant physical mechanisms in free shear layer flows. MZMD represents an accurate and efficient generalization of DMD and we demonstrate that MZ memory improves the ability to capture nonlinearities in flow and provide insights into the underlying mechanisms that lead to the generation of turbulence. We demonstrate that MZMD models enhance prediction of small- and large-scale flow features and allow refinement of these predictions via targeted manipulation of unstable modes. This leads to markedly improved agreement with ground truth across instantaneous fields and higher-order turbulence statistics, including production and dissipation rates, compared to standard DMD. This opens the door to identify modal couplings that drive the distribution of production and dissipation processes in the flow, enabling a clearer understanding of how coherent-structure dynamics influence turbulent energy pathways.

I. Introduction

Mixing layers belong to a class of fundamental shear-driven flows that are ubiquitous in natural and practical engineering systems. The scientific interest in mixing layers stems from their wide applicability. In both nature and practical applications, they govern a wide range of processes such as mixing within internal or jet combustion engine [1, 2], geophysical transport flows [3, 4], fuel-ablator mixing within internal confinement fusion reactors [5, 6], etc.

Mixing layers or free shear layers are broadly classified into two categories: single- and two-stream shear layers. A two-stream shear layer is formed upon an interaction of two fluid streams with differing streamwise velocities, while a single-stream shear layer (SSSL) emerges when one of the streams is effectively stagnant. The present work is focused on the latter class of free-shear layer. The characteristic inflectional velocity profile that defines such flows is inherently unstable and gives rise to the well-known Kelvin-Helmholtz (KH) instability. This instability initiates vortex roll-up leading to the formation of coherent vortical motions convecting downstream. These motions span a wide range of scales—from large, energy-containing motions down to the smallest dissipative eddies—convecting spatio-temporally with the coherent vortices. The mixing dynamics of these coherent motions are highly complex due to their inherently inhomogeneous and anisotropic nature—complexity that is further amplified in multi-fluid flows with varying physical

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properties. This complexity presents a significant challenge for both experimental characterization and numerical modeling. Despite over 75 years of focused research, our understanding of turbulent mixing dynamics, particularly the role of coherent structures and underlying physics behind their evolution and scale interactions, remains incomplete.

Modern flow field measurement techniques such as Particle Image Velocimetry (PIV) or Particle Tracking Velocimetry (PTV) generate large volumes of high-dimensional, nonlinear dynamical-systems data, demanding analysis that can distill the essential physics from measurement noise and incidental detail [7]. Modal decomposition has proven especially effective in this setting, isolating the coherent flow structures that dominate energy transfer and govern phenomena such as laminar–turbulent transition and shear layer mixing flows [8–12]. Because many canonical flows share recurring, low-rank features [13], these decompositions not only provide insights into the underlying mechanisms but also result in compact spectral representations for reduced-order prediction and control [12, 14].

Within this broader context, the Mori–Zwanzig (MZ) formalism, originally developed in statistical mechanics [15, 16], offers a mathematically rigorous alternative to Koopman formulation, providing nonlocal memory closures when a Koopman invariant subspace is unavailable [17, 18]. By projecting the Liouville equation onto a chosen observable subspace, MZ yields an exact Generalized Langevin Equation (GLE) whose Markovian term, memory kernel, and orthogonal dynamics jointly capture the influence of unresolved scales. This non-Markovian structure is particularly attractive for experimental shear-layer flows, where broadband turbulence, intermittent vortex merging, and sensor noise violate the assumptions underpinning memory-free models.

Recent work demonstrates that Mori’s linear projection enables fully data-driven identification of the MZ operators [17, 19–22]. Neural-network implementations [23] and their extensions to turbulent and transitional regimes [24–26] reveal that learned memory terms supply higher-order corrections, which can provide stabilizing and extending prediction horizons over Koopman operators [27]. Alternatives such as NARMAX remain equivalent to MZ only under a Wiener projection and typically assume Gaussian residuals [28], an assumption seldom met in shear-layer experiments. Other approaches, for example, memory-dependent DMD [29] or dynamic-MZ- τ LES modeling [30], highlight the growing interest in non-local closures while differing in projection choice and optimization strategy than the current approach.

This study utilizes the recently developed Mori–Zwanzig mode decomposition (MZMD) technique [18], which generalizes DMD to provide an accurate and efficient spectral representation of high dimensional nonlinear dynamics. MZMD includes MZ memory in order to account for the fact that the state space observables selected with DMD do not form a Koopman invariant subspace for nonlinear dynamics. Since nonlinear dynamics play a key role in shear layer mixing flows, MZMD provides a more expressive modal analysis technique. MZMD computes the modal decomposition of the generalized Langevin Equation (GLE), with the use of Mori’s projection. MZMD then quantifies the dominant memory contributions and can significantly improve to resolution of nonlinear dynamics compared to DMD [18]. This manuscript presents the application of the MZMD framework to experimentally acquired time-resolved data and examines how targeted manipulation of MZ modes enhances the prediction of large-to-small-scale flow features, demonstrating clear advantages over the standard DMD approach. After demonstrating the MZMD improves the ability to reconstruct and predict the shear layer states under question, we perform an analysis and interpretation of these MZMD modes and spectrum.

II. Methods

A. Mori-Zwanzig Mode Decomposition: Brief Outline

Following the works by Woodward et al. [18, 31], we now briefly describe the MZMD method and background. The Mori-Zwanzig formalism, introduced by Mori [15] and Zwanzig [16], offers a framework for deriving reduced-order models using observables that are experimentally measured or numerically simulated. Unlike Markovian methods such as DMD, it captures unresolved dynamics through non-Markovian memory terms and orthogonal dynamics [17]. As shown in Lin et al. [17], the formalism yields a generalized Langevin equation (GLE) for reduced observables $\mathbf{g}_{t+1} : \mathbb{R}^N \rightarrow \mathbb{R}^r$:

$$\mathbf{g}_{t+1} = \mathbf{\Omega}_0 \mathbf{g}_t + \sum_{k=1}^t \mathbf{\Omega}_k \mathbf{g}_{t-k} + \mathbf{W}_k \quad (1)$$

where $\mathbf{\Omega}_0$ is the Markovian term, $\mathbf{\Omega}_k$ are memory operators, and $\mathbf{W}_k$ represents orthogonal dynamics dependent on initial state $\mathbf{x}_0 \in \mathbb{R}^N$. The GLE (1) provides a closed and exact evolution equation (discrete form of an integro-differential equation) for the subspace of observables. As following [23], regression forms the projection operator. This projection operator is orthogonal to the orthogonal dynamics (as its defining feature $\mathcal{P}\mathbf{W} = 0$). With regression as the projection operator, the learned GLE can be projected on the space spanned by the observables, in which case the orthogonal dynamics is zeroed out. The learning algorithm incrementally minimizing the orthogonal dynamics so that the generalized fluctuation dissipation relation is satisfied. The GLE for state variables $\mathbf{x}_t$ and reduced observables $\mathbf{g}_t$ becomes:

$$\mathbf{x}_{t+1} = \mathbf{\Omega}_0^{(x)} \mathbf{x}_t + \mathbf{\Omega}_1^{(x)} \mathbf{x}_{t-1} + \cdots + \mathbf{\Omega}_k^{(x)} \mathbf{x}_{t-k}; \quad \mathbf{g}_{t+1} = \mathbf{\Omega}_0^{(g)} \mathbf{g}_t + \mathbf{\Omega}_1^{(g)} \mathbf{g}_{t-1} + \cdots + \mathbf{\Omega}_k^{(g)} \mathbf{g}_{t-k} \quad (2)$$

where (x) and (g) indicates operators in state and reduced space, respectively. The reduced observables are defined as $\mathbf{g}_t = \mathbf{U}_r^* \mathbf{x}_t$, where $\mathbf{X} = [\mathbf{x}_0 \mathbf{x}_1 \cdots \mathbf{x}_T] \approx \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^*$ via truncated Singular Value Decomposition (SVD) with projected memory operators: $\mathbf{\Omega}_k^{(g)} = \mathbf{U}_r^* \mathbf{\Omega}_k^{(x)} \mathbf{U}_r$ [18].

A snapshot matrix $\mathbf{G}$ is formed from $\mathbf{g}_t$ for compute two-time correlation $\mathbf{C}_i = \mathbf{G}_i \mathbf{G}_0^T$ for $i = 0 \rightarrow k$. The Markovian term is given by $\mathbf{\Omega}_0^{(g)} = \mathbf{C}_1 \mathbf{C}_0^{-1}$ and memory terms are recursively obtained via:

$$\mathbf{\Omega}_i^{(g)} = \left(\mathbf{C}(i+1) - \sum_{l=0}^{k-1} \mathbf{\Omega}_l^{(g)} \mathbf{C}(i-l) \right) \mathbf{C}^{-1}(0) \quad (3)$$

Memory coefficients are used to construct the companion matrix:

$$\mathbf{C}_g = \begin{bmatrix} \mathbf{\Omega}_0^{(g)} & \mathbf{\Omega}_1^{(g)} & \cdots & \mathbf{\Omega}_k^{(g)} \\ \mathbf{I} & \mathbf{0} & \cdots & \mathbf{0} \\ & & \ddots & \\ \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} \end{bmatrix} \quad (4)$$

The reduced dynamics are given by: $\mathbf{g}_t = \mathbf{P}_0 \mathbf{C}_g^t \tilde{\mathbf{z}}_0$, $\tilde{\mathbf{z}}_0 = [\mathbf{g}_0 \mathbf{g}_1 \cdots \mathbf{g}_{-k}]^T$, $\mathbf{P}_0 = [\mathbf{I} \mathbf{0} \cdots \mathbf{0}]$ with eigenpairs $(\lambda, \mathbf{w})$ of companion matrix $\mathbf{C}_g$ characterizing the system. The full-state solution

is: $\mathbf{x}_t = \sum_i a_i \lambda_i^t \boldsymbol{\phi}_i^0$, $\boldsymbol{\phi}_i^0 = \mathbf{P}_0 \boldsymbol{\phi}_i$. These modes are computed efficiently and related to the corresponding block companion matrix formed from $\boldsymbol{\Omega}_i^{(x)}$'s. These are related as follows: Let $(\lambda, \mathbf{w})$ be an eigenpair of $\mathbf{C}_g$. Then $(\lambda, \boldsymbol{\psi})$ is an eigenpair of $\mathbf{C}_x$ where $\boldsymbol{\psi} = [\boldsymbol{\psi}^0, (1/\lambda)\boldsymbol{\psi}^0, \dots, (1/\lambda^k)\boldsymbol{\psi}^0]^T$, and $\boldsymbol{\psi}^0 = \mathbf{U}\mathbf{w}^0$.

One may encounter unstable eigenvalues that can cause the prediction fields to diverge, a point discussed in greater detail in later sections of this manuscript. To address this, two stabilization strategies are employed:

- 1) excluding all eigenvalues with $|\lambda_i| > 1$, and
- 2) projecting the unstable eigenvalues onto the unit circle, yielding stabilized modes as defined by

$$\lambda_i^* = \frac{\lambda_i}{\sqrt{(1 + (\Im(\lambda)/\Re(\lambda))^2)}} \quad (5)$$

where λ_i^* denotes the stabilized eigenvalue.

The MZ models constructed out of these two approaches are named as MZMDEx and MZMDSt, respectively.

B. Turbulent-kinetic-energy (TKE) budget

The mixing dynamics of shear layer lies in the understanding of TKE budget terms, especially production and dissipation rates. For an incompressible, constant-density flow the turbulent-kinetic-energy $k = \frac{1}{2} \overline{u'_i u'_i}$ obeys

$$\frac{\partial k}{\partial t} + U_j \frac{\partial k}{\partial x_j} = \underbrace{-\overline{u'_i u'_j} \frac{\partial U_i}{\partial x_j}}_P - \frac{\partial}{\partial x_j} \left(\frac{1}{2} \overline{u'_i u'_i u'_j} + \frac{\overline{p' u'_j}}{\rho} - \nu \frac{\partial k}{\partial x_j} \right) - \underbrace{\nu \frac{\partial u'_i}{\partial x_j} \frac{\partial u'_i}{\partial x_j}}_\varepsilon. \quad (6)$$

where U_j is mean velocity. In the self-similar region of a statistically stationary planar mixing layer the diffusion terms are small, thus there is an effective balance between production and dissipation rates, $P \approx \varepsilon$.

C. Non-orthogonal modal decomposition

Velocity fluctuations are written as a truncated expansion in generally *non-orthogonal* basis functions with MZMD

$$\mathbf{u}'(\mathbf{x}, t) = \sum_{j=1}^N a_j(t) \boldsymbol{\phi}_j(\mathbf{x}). \quad (7)$$

Substituting this into TKE equation 6 to compute the dominant mode interactions that leads to production and dissipation.

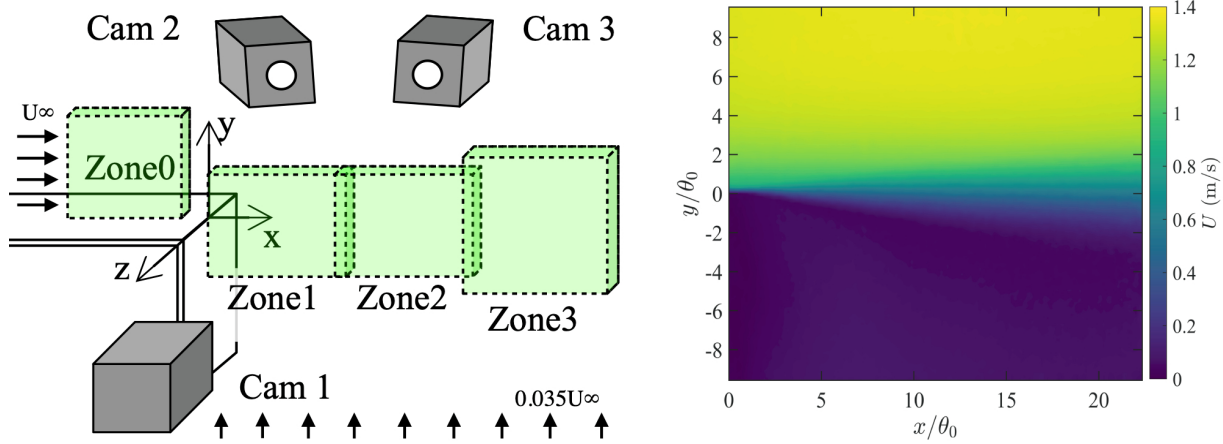


Figure 1 (Left) Schematic of experimental and flow acquisition setup. (Right) Mean velocity contour field in Zone1 (spanwise-averaged).

III. Single-stream shear layer: 4D-PTV measurements

A high-Reynolds number water tunnel facility at Michigan State University specifically constructed to produce single-stream shear layers (SSSL) is utilized. An schematic of the experimental setup is shown in Figure 1a; the incoming boundary layer (BL) develops in the tunnel which subsequently separates at the 90° backward facing separation edge, forming an SSSL. A time-resolved three-dimensional Particle Tracking Velocimetry (PTV) technique captures the Lagrangian flow trajectories within shear layer over the flow volumes named as Zone1-3. The Lagrangian trajectories are transformed onto a structured grid to obtain Eulerian flow fields; refer to [32] for more details. The current work is only concerned with the flow field data obtained over Zone1, immediate to the separation, acquired at a repetition rate of 4 kHz for a period of 1 s amounting to 4 000 snapshots [32]. The instantaneous 3D-flow fields are spanwise-averaged, considering the spanwise homogeneity of a planar shear layer. In addition, the incoming flow conditions measured in the boundary layer region (marked as Zone0; refer Figure 1a) are tabulated in Table 1.

Flow parameter	Value
Free-stream velocity, U_∞ (m/s)	1.37
Friction velocity, u_τ (m/s)	0.0532
BL thickness, $\delta_{99\%}$ (mm)	23.4
Momentum thickness, θ_0 (mm)	1.88
$Re_\tau = u_\tau \delta_{99\%} / \nu$	1 240
$Re_\theta = U_\infty \theta_0 / \nu$	2 566

Table 1 Incoming flow parameters estimated in the boundary layer (BL) region, marked as Zone0 in Figure 1.

IV. Results and Discussion

The POD mode energy variation across training dataset reveals that approximately 90%, 95% and 99% of the energy is contained for ranks (r) 171, 486 and 1478, respectively, refer to Figure 2. A relatively large variation among the marked ranks portrays that an ample amount of low-rank structures are present in the flow and comparatively larger amount of “intermediate”-rank structures represent the flow system immediate downstream to the separation edge. Consequently, the rank r values of 200, 500 and 1500 are used for reduced-rank assessment of the flow captured in the region.

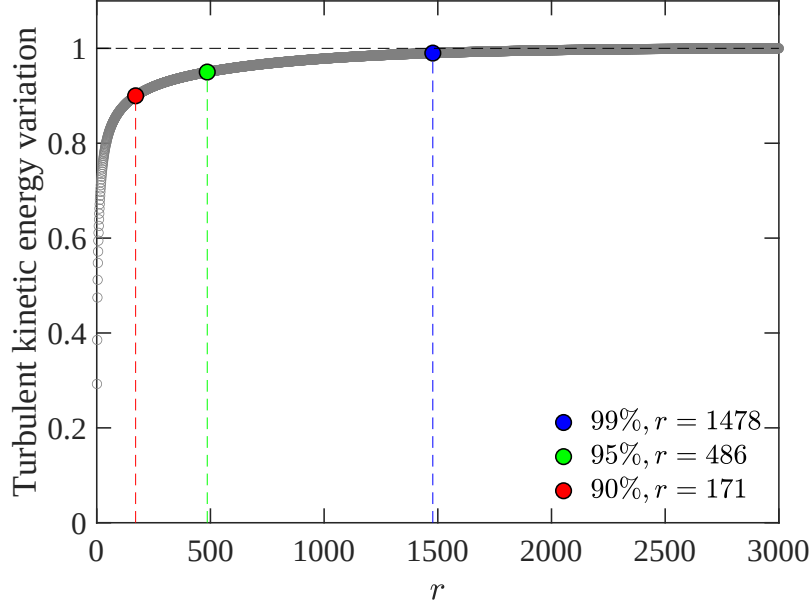


Figure 2 Turbulent kinetic energy variance in POD modes.

Figure 3 presents the evolution of the Frobenius norm of the memory operators for $r = 200, 500$, and 1500 , with memory terms up to $k = 15$, evaluated across the available snapshots. The goal is to determine how many snapshots are sufficient for the operators to converge reasonably. For $r = 200$, all memory terms exhibit a clear tendency to level off, indicating convergence. The results for $r = 500$ and $r = 1500$ also show plateau-like behavior, though the magnitude and location of these plateaus differ. In this work, we select $r = 500$ for further analysis; the further rationale for this choice shall become much clearer in later sections. Briefly, $r = 200$ captures only the largest energetic scales, which leads to a more apparent early plateau and very small operator norms for memory terms with $k \geq 2$. In contrast, for $r = 1500$ a broader range of flow scales are expected to be captured, the available snapshots may be insufficient for the memory operators to fully converge. Nonetheless, a total of 3 500 spanwise-averaged time-resolved 2D-flow field data-frames are used as training dataset, followed by subsequent 200-frames gap for establishing initial condition, and remaining 300 time-steps withheld for model validation.

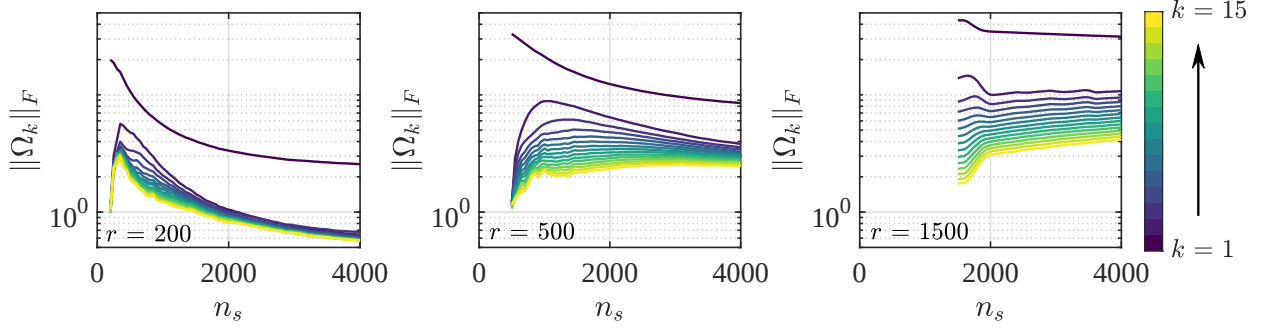


Figure 3 Evolution of Frobenius norm of the memory operators corresponding to $k = 1$ to 15, for $r = 200, 500$, and 1500 over the captured snapshots.

A. Eigenvalue spectrum: DMD and MZMD

Figure 4 presents the eigenvalues and corresponding spectra for $r = 500$ and $k = 0, 2, 6$ and 10. The eigenvalues are color-coded to highlight their associated amplitudes in the spectral plot. The $k = 0$ case corresponds to standard DMD, while increasing k introduces additional decaying modes, visible as clusters of eigenvalues within the unit circle. A striking trend emerges: as more memory terms are incorporated, an additional circular band of modes appears, representing a distinct ‘memory mode’ that represents a distinct decay mode that is expected to capture features of the flow dynamics that standard DMD fails to capture. We anticipate that adding further memory terms would continue to generate additional modal bands, thereby enhancing prediction accuracy. Moreover, these decaying modes have increasingly larger eigenvalue magnitudes with increasing k . Although, a majority of these decaying modes correspond to low amplitude, the collective contribution of these modes is expected to be significant towards prediction of the faster flow scales within the region.

Figure 5 shows the eigenvalues and spectra for $r = 500$ and $k = 10$. Note that several of the modes with significant spectral amplitude are highlighted; their corresponding eigenvalues lie outside the unit circle, indicating potential instability. Including these modes in the model is expected to eventually cause divergence in the predicted fields. To address this, two stabilization strategies are considered, first, removing the unstable modes altogether (named as MZMDEx), second, projecting their eigenvalues onto the unit circle (named as MZMDSt). Alongside the full model (named as MZMD), these two modified models are constructed (as discussed in section II.A) for comparative evaluation.

B. Comparing Model Predictions

The prediction signal are obtained for 300–time steps, corresponding to a total of 75 ms, approximately equal to time-steps ~ 4 to 14 times of the Kolmogorov time-scale, $\overline{\tau}_\eta$ ($= 12 \pm 6.5$ ms), where $\overline{\tau}_\eta$ is obtained across the shear layer captured in Zone1.

1. Velocity field

Figure 6 shows the predicted velocity (streamwise and cross-stream) signals for $r = 500$ obtained from various MZ models, compared against the ground truth and the standard DMD prediction. The signal is sampled at a point located at the center of the shear layer. The DMD prediction shown here

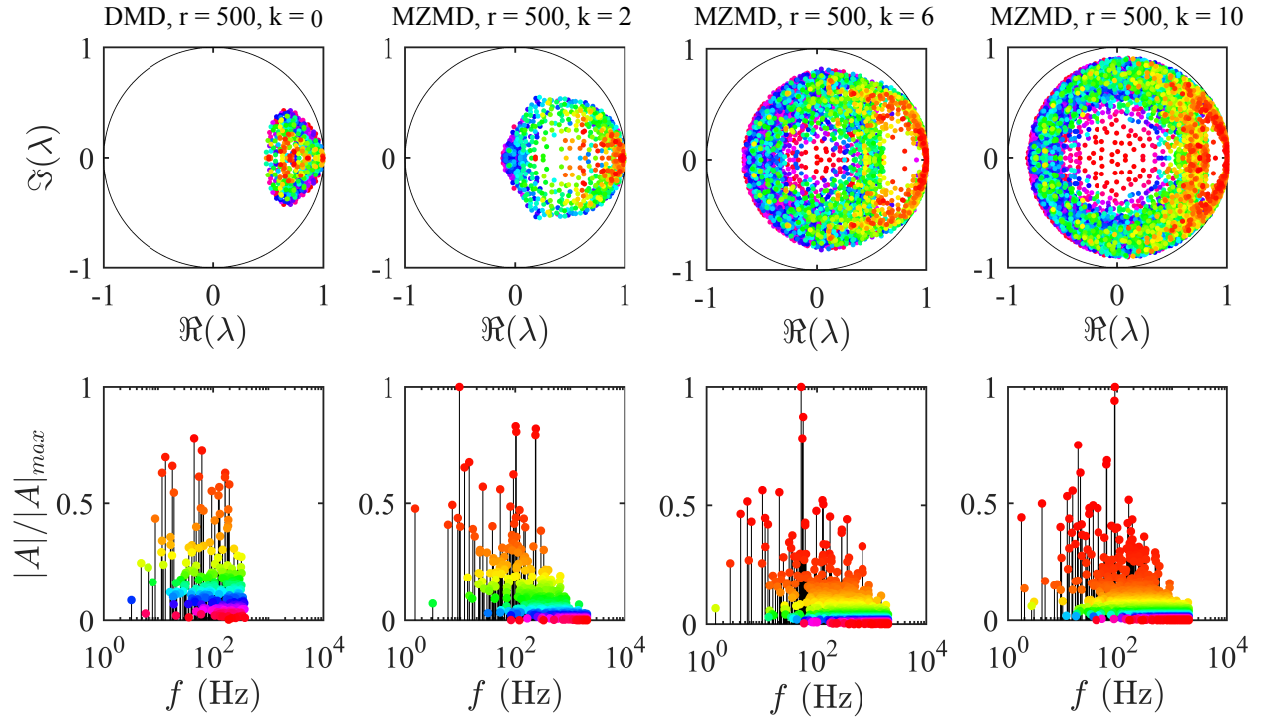


Figure 4 Eigenvalues (top row) color-coded correspondingly to the frequency spectrum (bottom row) for $r = 500$ and $k = 0, 2, 6$ and 10 . Here $k = 0$ corresponds to standard DMD.

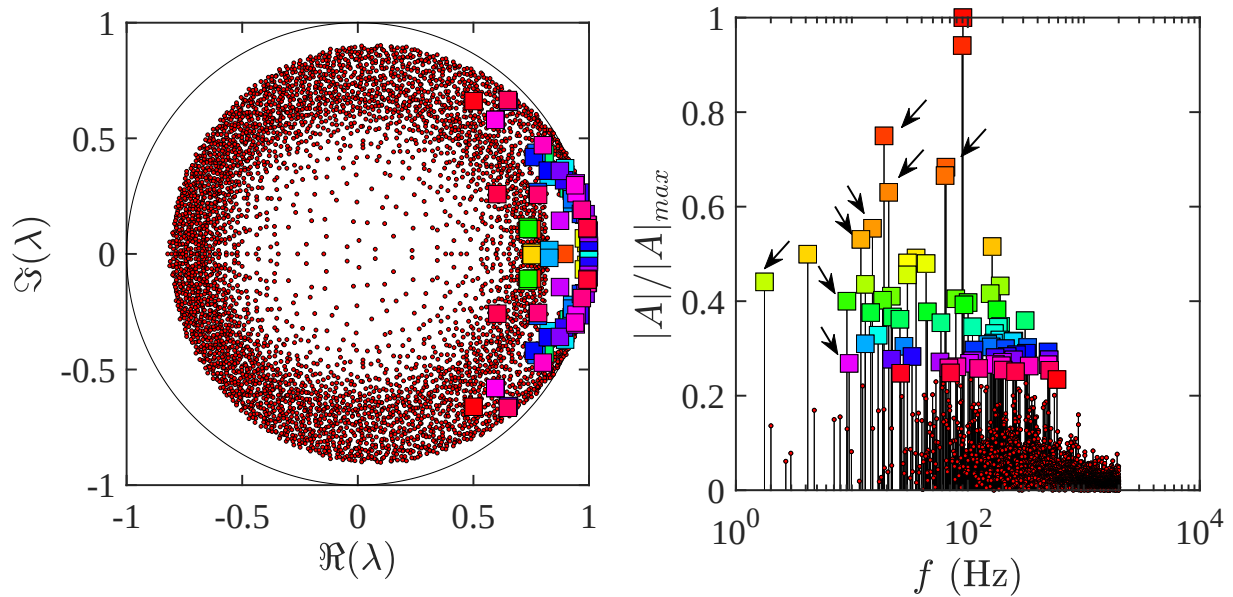


Figure 5 MZMD eigenmodes and spectrum for $r = 500$ and $k = 10$. The amplitude–frequency plot highlights selected modes (color-coded correspondingly to eigenmodes), with black arrows indicating several unstable, high-amplitude modes.

captures only the dominant large-scale energetic structures and quickly decays, failing to reproduce the higher-frequency, fast-scale fluctuations. In contrast, the MZMD model provides a more faithful reconstruction of the flow features, including partial recovery of the higher-frequency content, although its predictions exhibit noticeable magnitude deviations or divergence from the ground truth at later times. The stabilized variant (refer to II.A), MZMDSt, maintains signal stability over longer prediction horizons; however, a phase offset, at later time, relative to the ground truth becomes evident. Notably, the MZMDEx model (refer to II.A), in which unstable modes are removed, remains stable throughout the full prediction window and closely tracks the ground-truth signal. Both small- and large-amplitude fluctuation features are reproduced with remarkable accuracy in this case.

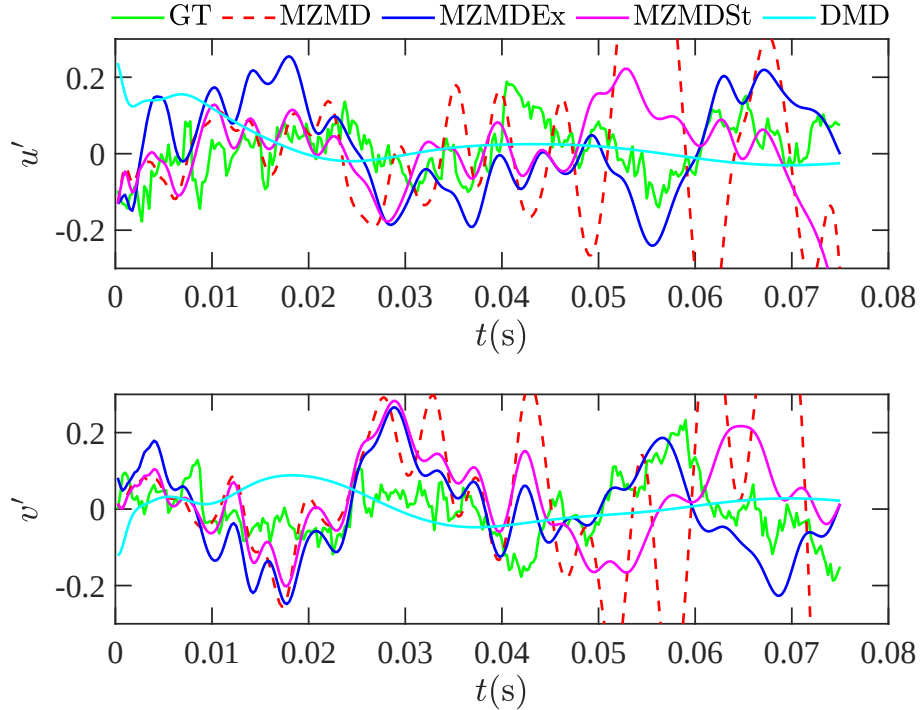


Figure 6 Comparison of predicted velocity (in m/s) signals from different models at a centerline location in the SSSL for $r = 500$ and $k = 10$, evaluated against the ground truth (GT).

Furthermore, the contours of the velocity fluctuation fields for both components are presented for the “short-time” prediction ($t_s = 50$) and the “long-time” prediction ($t_s = 250$) in Figures 7 and 8, respectively. For short-time predictions, the MZ-based models reproduce the flow structures with similar spatial features and comparable magnitudes, while the DMD results capture only the dominant large-scale motions. The finer-scale fluctuations are represented far more effectively by the MZ models, and this behavior remains consistent across the variants. At longer prediction times, the MZ models continue to retain much of the fine-scale structure, although their predicted pointwise magnitudes begin to deviate from one another. As expected, the DMD solution rapidly dissipates, failing to preserve either the fine-scale features or the overall fluctuation magnitudes.

Further assessment of the predicted turbulent fields is provided through the probability distribution

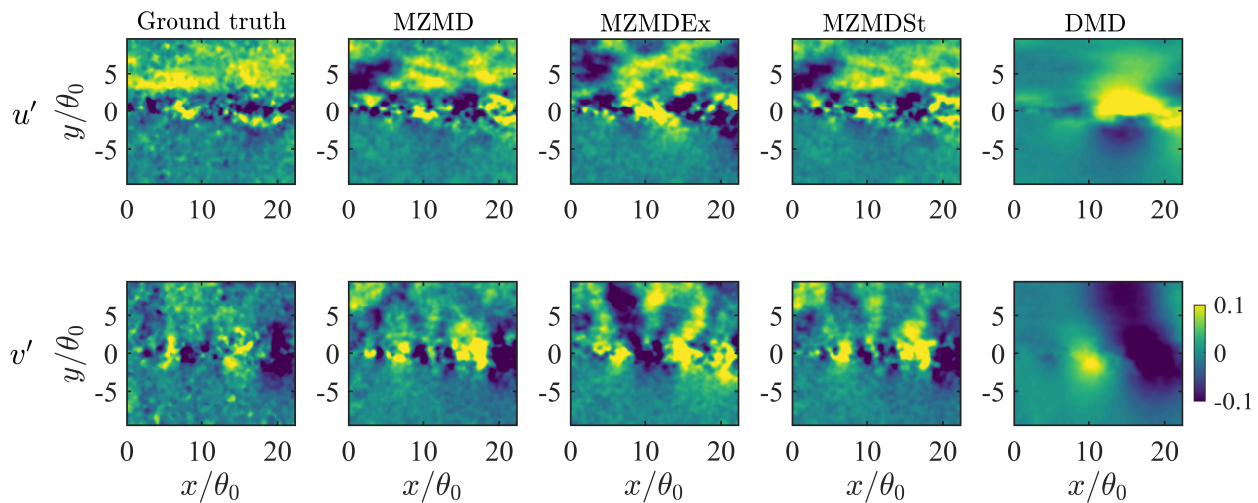


Figure 7 Predicted velocity (in m/s) contours across different models compared against ground truth at a time-step $t_s = 50$ i.e. $t = 12.5$ ms after the initial condition.

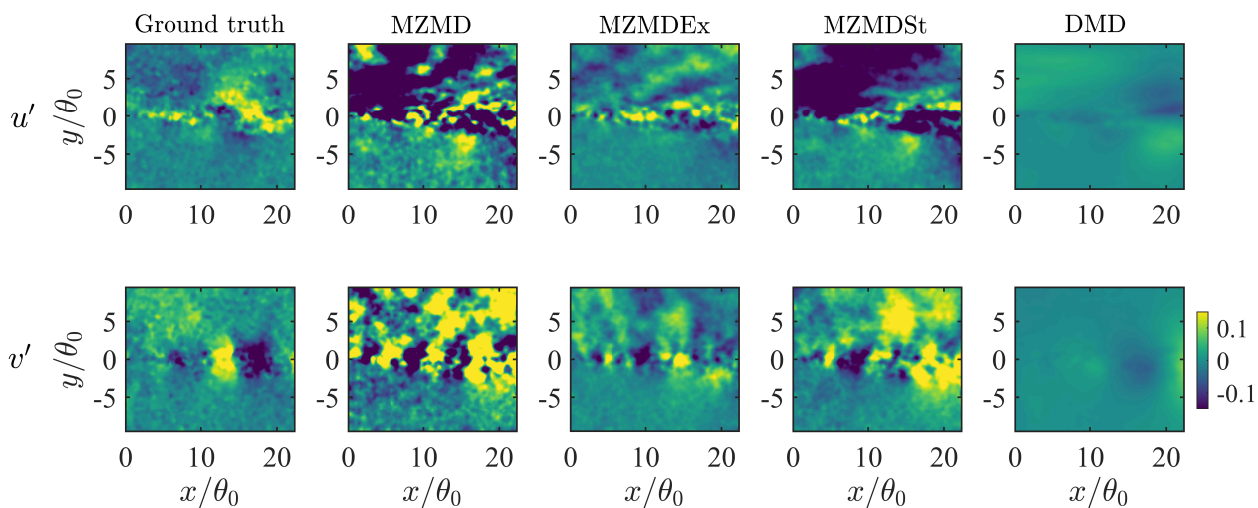


Figure 8 Predicted velocity (in m/s) contours across different models compared against ground truth at a time-step $t_s = 250$ i.e. $t = 62.5$ ms after the initial condition.

functions (PDFs) of the velocity fluctuations for each model compared with the ground truth. To distinguish between short- and long-time predictions, PDFs are computed using the first 150 predicted full flow fields for the short-time window and the subsequent 150 fields for the long-time window. The DMD results exhibit noticeably thinner tails, consistent with its previously noted inability to represent the smaller, fast-scale fluctuations whose signatures appear as broader tails in the ground-truth distribution. In contrast, the MZ-based models closely match the ground-truth PDFs near the core region of small velocity fluctuations, although they generally produce somewhat broader tails. The significantly broader tail observed for the MZMD model reflects the divergence in its prediction at later times. The remaining MZ variants show better agreement, capturing the tail behavior more faithfully. Importantly, the broader tails predicted by the MZ models indicate their ability to reproduce intermittency and the associated large-amplitude fluctuation events within the flow.

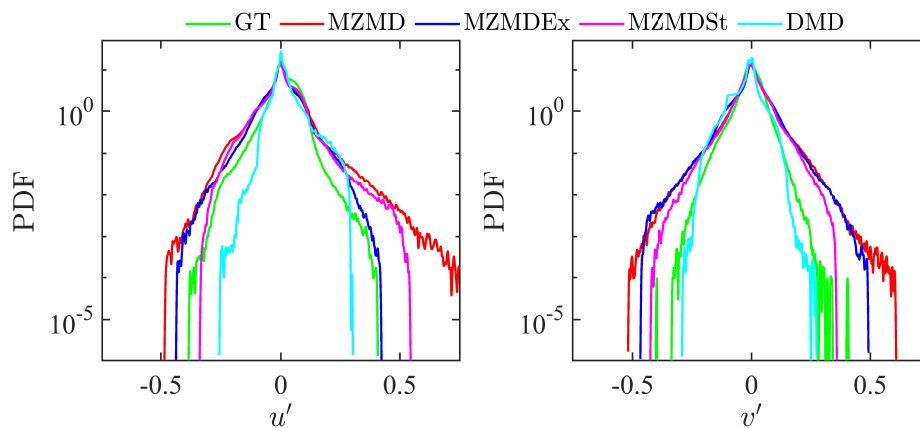


Figure 9 PDFs of ensemble velocity prediction data obtained for entire field over early time steps spanning from $1 \leq t_s \leq 150$ for different models.

For the long-time statistics, the PDFs for the MZMD and MZMDSt models exhibit substantially broader tails, reflecting both the divergence of the MZMD predictions and the phase lag with respect to ground truth observed in the MZMDSt velocity signals. In contrast, the MZMDEx model remains stable throughout the prediction window, with PDF tails that closely resemble those of the ground truth, reaffirming its ability to closely reproduce the turbulent field statistics. The variability across the different MZ models highlights the flexibility of the MZ decomposition framework, demonstrating how model characteristics can be tuned to better match the underlying flow physics. As expected, the DMD predictions degrade further at longer times due to its lack of memory retention, limiting its ability to sustain accurate representations of the turbulent dynamics compared to MZ models.

2. Reynolds shear stress fields

In turbulence modeling, the Reynolds shear stress is a key quantity in Reynolds-averaged approaches as an appropriate closure for it is required. Therefore, we compute the Reynolds shear stress fields from the predicted flow fields of each model for comparisons. The contour plots for these fields at early and later times are shown in Figure 11. As Reynolds shear stress primarily represents

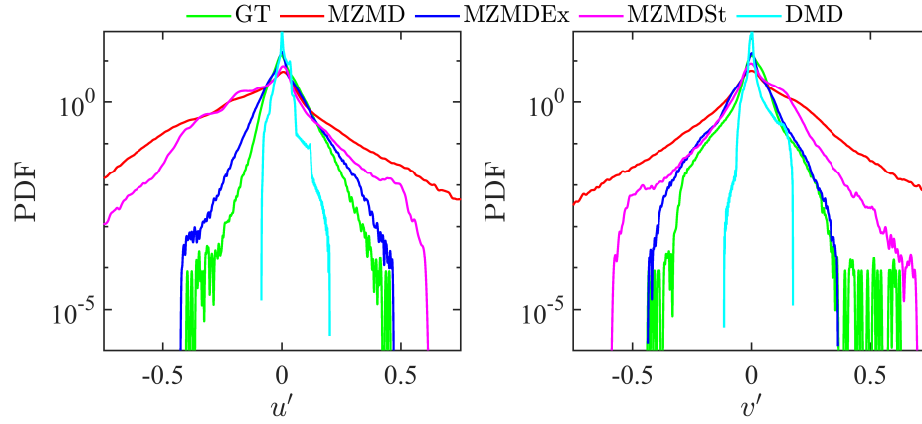


Figure 10 PDFs of ensemble velocity prediction data obtained for entire field over later time steps spanning from $151 \leq t_s \leq 300$.

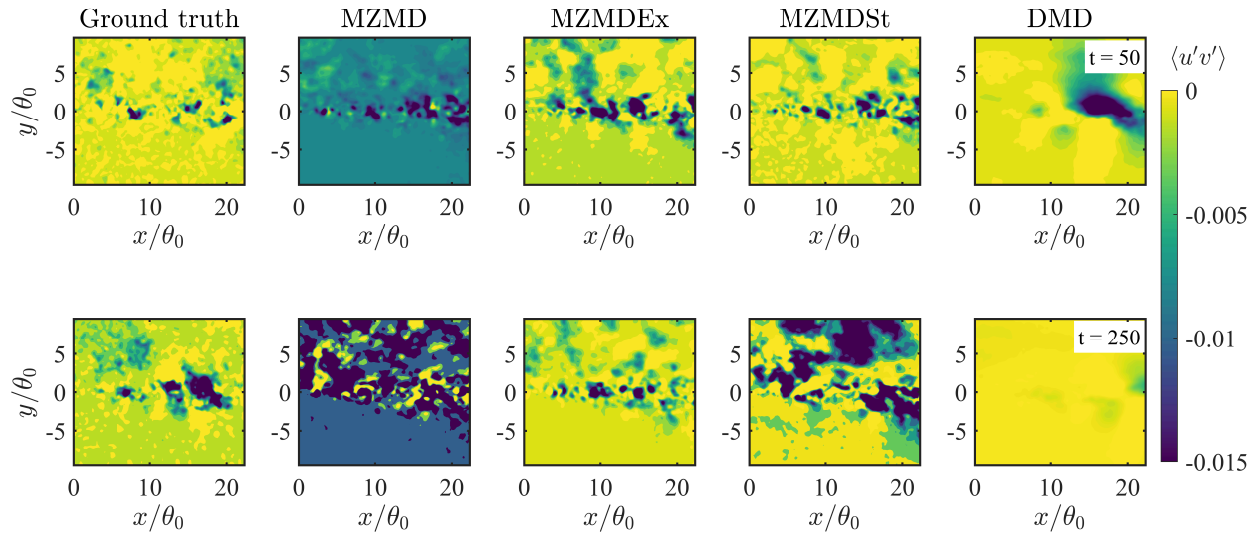


Figure 11 Comparison of predicted Reynolds shear stress, $\langle u'v' \rangle$ (in m^2/s^2) contours across different models at time-step $t_s = 50$ and 250 .

smaller-scale turbulent motions, the predictions are generally less accurate than those of the velocity fields. Nevertheless, for all the MZ-based models, essential flow features remain discernible at early times with expected divergence and larger magnitudes at later-times. As anticipated, the DMD predictions fail to capture these fine-scale stress features altogether.

The mean Reynolds shear stress fields derived from the predicted datasets (Figure 12) capture shear-layer growth patterns that closely follow the ground truth. There still remains a discrepancy in magnitude, nonetheless, this highlights the importance of accurately representing modal interactions between the streamwise and cross-stream velocity components in predicting growth dynamics of a shear layer.

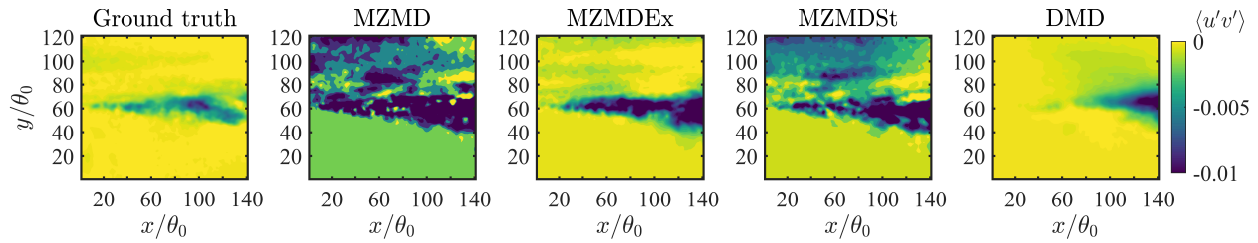


Figure 12 Mean Reynolds shear stress, $\langle u'v' \rangle$ (in m^2/s^2) contours obtained over the prediction time-set.

3. Production and Dissipation Rates

We further evaluate model performance by examining the TKE budget terms, specifically, the production and dissipation rates, using the contour fields shown in Figures 13 and 14, respectively. Consistent with earlier observations for other flow quantities, the MZ-based models remain reasonably close to the ground truth even at later prediction times. As expected for higher-order energetic quantities, the predicted features do exhibit difference in magnitudes across the MZ models. Nevertheless, a key result is that the MZ models successfully retain memory effects, enabling them to reproduce dynamics that extend into the small-scale regime, an ability that marks a significant improvement over standard DMD and its common variants. The broader motivation for employing the MZ decomposition here is to potentially identify dominant mode couplings and their contribution towards total production and dissipation rates; and the results indicate clear promise in this direction. Future work will incorporate the non-orthogonal velocity representation (equation 7) into the TKE budget framework (equation 6) to directly quantify these modal couplings governing the energy transfer and distribution within the coherent structures.

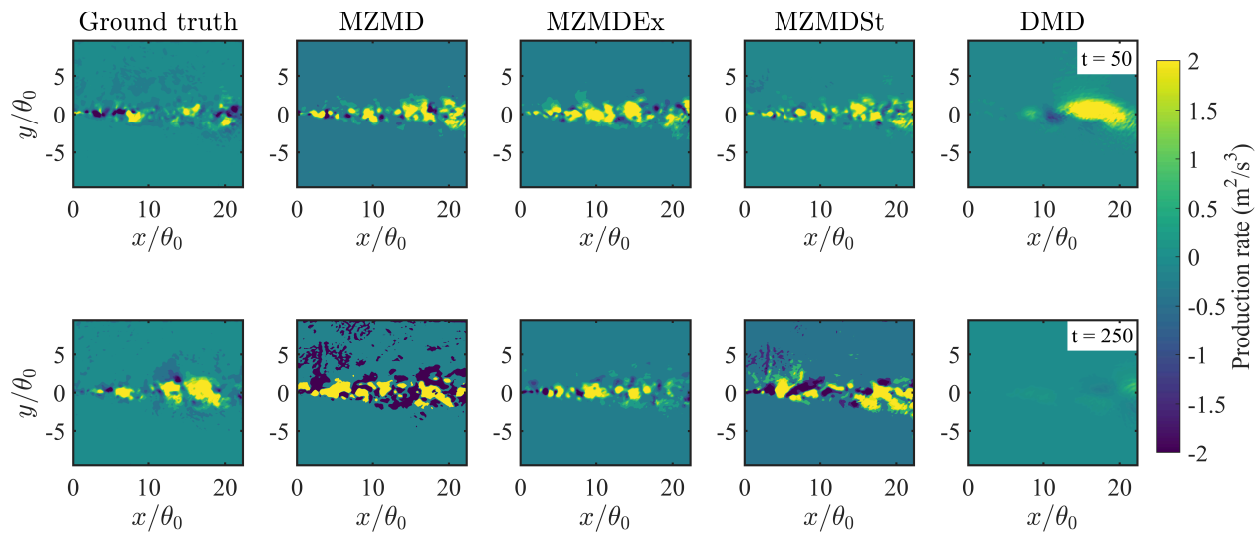


Figure 13 Comparison of predicted production rate (in m^2/s^3) contours across different models at time-step $t_s = 50$ and 250.

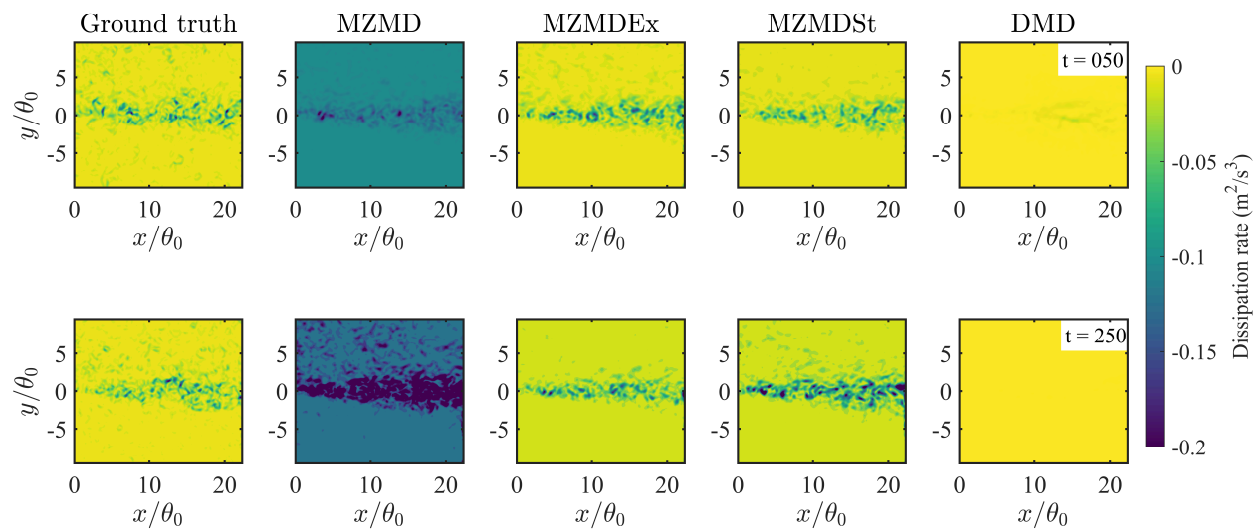


Figure 14 Comparison of predicted dissipation rate (in m^2/s^3) contours across different models at time-step $t_s = 50$ and 250.

V. Conclusions and Future Directions

In this manuscript, we apply the Mori–Zwanzig Modal Decomposition (MZMD) to time-resolved velocimetry measurements acquired in a single-stream shear layer. The results demonstrate that MZ-based models, through their incorporation of memory terms, offer substantially improved predictive capability relative to standard DMD. In particular, the MZ framework captures a broader range of flow scales and preserves coherent structures over longer time scale, as reflected in both the predicted velocity signals and its probability distribution functions. Thus, a key takeaway is that the MZ memory terms, acting as an approximate closure to DMD, enhance the resolution of small-scale structures and improve the ability to capture dynamics associated with the nonlinearities of the flow.

A detailed examination of the resulting MZ modes further highlights the flexibility of the framework: by selectively modifying (stabilizing), or excluding, unstable modes, one can stabilize long-time predictions, thus influencing error growth and divergence behavior. The predicted flow fields and derived higher-order turbulence quantities, including Reynolds shear stress, TKE production, and dissipation rates, exhibit markedly better agreement with the ground truth than those obtained from standard DMD or its variants. Importantly, the MZ models more faithfully reproduce small-scale dynamics, yielding improved fidelity in both instantaneous and statistical representations of the shear layer.

These findings underscore the utility of MZMD as a data-driven reduced-order modeling tool capable of embedding physically relevant memory effects, capturing nonlinear modal interactions, and offering interpretable levers for stabilizing predictions. Future efforts will center on how modal couplings govern the production–dissipation balance within coherent structures, where memory effects likely play an amplified role.

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Declaration of interests. The authors report no conflict of interest.

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